Promoting Local Revitalization to Solve Issues on Degraded Forests in Japan: A Case Study in Oku-Aizu, Fukushima

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Abstract

The degradation of forest ecosystems due to depopulation, aging and economic depression of industries in rural areas, has been extensively discussed, while at the same time, increased demand for renewable energy from woody biomass has resulted in much attention to forestry and rural areas. This study provides an overview of regional circulation in a small woody biomass energy system in a typical Japanese rural area, Mishima Town, Fukushima Prefecture. To resolve the problem of forest degradation, two issues (energy and forests) in the study area are focused on and discussed in light of the national-level background. We have derived two constraints (circulation radius and feasible size) and conducted four research activities (focused respectively on forest resources, production, energy infrastructure and the social system and scenarios) for local revitalization. To address implementation of an energy system, Mishima Town established a council that has working groups strongly related to our research activities. Finally, it can be stressed that the goal of our project is identical to the concept of "regional circular and ecological spheres" (regional CESs).

Key words : cogeneration system, forest management, regional circulation, renewable energy, woody biomass

1. Introduction

The degradation of forest ecosystems has been extensively discussed in Japan. This degradation can be attributed to decreasing population in rural areas and a decline in local industrial activity, as well as to synergistic effects arising from these factors. Meanwhile, power generation from woody biomass has gained attention because of the renewable energy feed-in tariff (FIT) system implemented in 2011 after Japan's nuclear disaster (METI, 2017). Increased demand for woody biomass can result in higher biomass production and promote forest harvesting. The latter includes forest carbon sinks and sites where carbon dioxide is absorbed, leading to climate change mitigation and implementation of policies to substitute biomass fuel for fossil fuels.

However, the FIT system is based on large biomass power plants, which may require an annual supply in the tens of thousands of tons of woody-biomass (Sakai et al., 2017). This implies that biomass procurement from local forests to produce fuel should require sustainable measures assessed through planned deforestation. Although FIT is a policy aiming to promote the development and installation of renewable energy at the initial stage, the permanence of high purchase prices compared to the price of electricity from conventional electric power companies is not guaranteed. After 2017, new solar power-generation facilities of 2,000 kW production or higher were no longer eligible for FITs (and that number was lowered to 500 kW in 2019). Although future power plants will need to operate on wholesale and retail schemes, it is difficult to implement them in large-scale domestic woody-biomass power plants. Thus, almost all of them will need to expect conditions of reliance on use of the FIT system.

This study addresses general issues related to Japanese forests and describes the efforts that have been conducted to achieve local revitalization and provide solutions for environmental problems. Subsequently, it discusses the adoption of renewable energy in Japan within the context of woody biomass and explores socioeconomic constraints on the use of forest resources as a modern energy source. Then it provides an overview of the regional circulation in a small combined heat and power (CHP) woody biomass energy system, which is the goal of this study. It describes the initiatives proposed in Mishima Town, Fukushima Prefecture, and their current results. Finally, it addresses the convergence of these initiatives with similar ones, such as the Regional Circular and Ecological Spheres (CESs) and Sustainable Development Goals (SDGs).

2. Issues Affecting Forests in Japan

Due to a lack of timber for construction in the 1960s, Japan made a major transition from firewood and natural forests to plantation forests. However, timber importation regulations were relaxed and price competition arose between domestic and imported timber. As a result, timber prices decreased, resulting in difficulties for domestic forestry industries. Labor forces have declined due to aging, and there has been a population decrease in rural and mountainous areas; thus a shortage of workers able to manage forests has arisen. Moreover, some plantation forests were not well managed before the FIT system was introduced for timber production. The management rate of these forests was approximately 45% (Amano, 2007).

Forests have various ecological functions, and their benefits to human society have ecosystemic importance (ecosystem services). The functions of forest ecosystems include prevention of sediment discharge, headwater conservation, water purification, carbon absorption, timber and forest product supply, and biodiversity conservation. These functional issues are affected (quantitatively and qualitatively) by forest management and the degree of forest conservation. Well-managed forest ecosystems exhibit higher carbon absorption rates (Hiroshima & Nakajima, 2006) and improved photosynthesis rates after thinning (Han & Chiba, 2009). In contrast, soils in unmanaged forests have lower water-holding capabilities and exhibit erosion (Miyata et al., 2009).

Those conditions currently prevail because of the increased demand for biomass power plants. Because clear-cutting is simpler to implement than thinning, and reforestation is difficult due to economic and labor factors, resource depletion could potentially occur due to the lack of adequate systems for achieving a managed degree of logging and reforestation in the affected forests. In addition to the amount of resources available, the effects of ecosystem functions have major impacts on environmental issues such as disaster prevention.

In 2019, the Forestry Agency launched the New Forest Management System, which established distinctions between forest ownership and forest management, thereby encouraging harvesting by operators interested in promoting municipal forest management (Forestry Agency of Japan, 2020; Uchiyama & Kohsaka, 2020). Currently, an economic and sustainable framework for forestry at the municipality level needs to be developed as soon as possible, considering the advantages of that system.

3. Renewable Energy and Regional Circulation

3.1 Renewable Energy in Japan

After the first oil crisis (1973), Japan's dependence on foreign energy sources became evident along with environmental issues due to the use of fossil fuels (air pollution and greenhouse gas emissions). Thus development of novel energy sources was initiated, aiming to solve those problems. "New energy" is a term used in Japan in a unique context because it refers to non-fossil fuels under the Act on the Promotion of New Energy Usage (New Energy Act). Currently, the definition by law is similar to that of the renewable energy concept, including biomass, solar thermal energy, snow and ice thermal energy, along with geothermal, wind and solar power generation.

The Great East Japan Earthquake and consequent nuclear disaster in 2011 became an opportunity for Japan to reconsider its energy sources. In addition to the existing trend in "fossil-free" sources for international security reasons, Japan considered renewable energy at that time as a means of decentralizing energy and managing local resources well autonomously. In addition, international treaties related to climate change have progressed, and promoting renewable energy is consistent with global agreements to phase out carbon energy and reduce future climate risks, according to the Paris Agreement. Although solar power was initially considered as a potential source of renewable energy, power generation from un-utilized woody resources has attracted attention, as it also maintains a relatively high purchasing price under the current system.

3.2 Regional Use of Woody Biomass and its Constraints

Historically, woody biomass has been used not only as construction material but also as an energy source. In Japan, firewood and charcoal were widely used as energy sources before households turned to fossil fuels in the first half of the 20_{th} century (Totman, 1995). Additionally, firewood forests were located around populated areas. Currently, however, profits from woody biomass as an energy source have not returned to their earlier values.

Households need energy for a variety of purposes and high amounts of energy are consumed. In the past, the energy demand in Japan arose primarily for warmth, hot water, and cooking. Nowadays, there is also demand for cooling, equipment operation and transportation, which are powered by electricity and fossil fuels (diesel fuel and gasoline). Consequently, energy demand in the past was low compared to current requirements. For instance, the annual electricity consumption in Fukushima Prefecture is $17{,}207 \times 10^6$ kWh (as per the 2010 record before the earthquake), which represents 6.757×10^6 tons of woody biomass in terms of heat demand. That amount is approximately four times the joint timber volume from plantation forests in the five towns and villages around Mishima. Togawa et al. (2019) also indicate such large demands (14,000 tons/y in the case of maximum use of woody biomass) that is supplied only to Mishima compared to the potential supply of woody biomass (4,000 tons/y). Converting that volume of resources into energy would lead to forest depletion in less than three months. In contrast, approximately 40 years would be required before harvesting a standard cedar forest.

One difficulty with woody biomass energy applications is the geographical limitation caused by the need for transporting materials from the supply area to the production area. To benefit from regional resources, including forest reserves, we should consider a "circulation radius" (Fig. 1) that is socioeconomically viable. The radius has been studied in the context of waste collection areas (e.g., Fig. 5 in Ohnishi et al., 2016). Although the present study area has an abundance of forest resources, there is a maximum limit regarding transportation distances of raw biomass corresponding to 50–125 km based on previous studies (e.g., for large-scale power plants in Hokkaido, Sakai et al., 2017). Furthermore, when a resource is considered an energy source, the range that electricity generated from it can reach using self-operated power lines is limited to a few kilometers due to economic constraints (Agency for Natural Resources and Energy, 2019; e.g., for case studies, Shima & Nose, 2014; Haruta & Muraki, 2017).

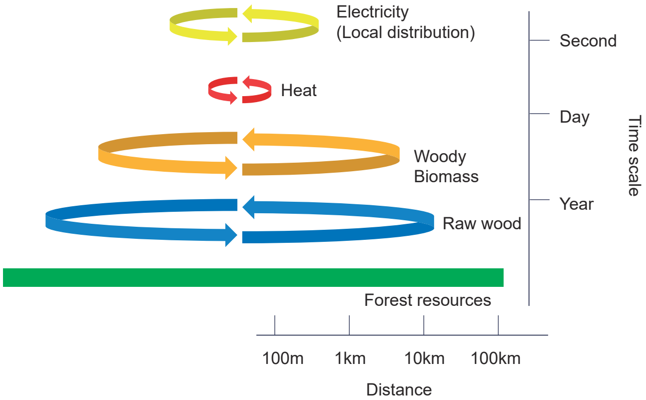

Fig. 1 Circular radius (geographical) and life span (temporal) of local resources (see text).

Similarly, the supply range when using self-operated pipes for thermal energy is a few hundred meters. Regarding electricity, it is possible to transport power generated by a small local power plant through a power grid owned by a regional power company that operates across a larger area. Grid-use fees, however, would be included in the retail electricity price, which would not be economical for consumers. Obtaining profits from regional resources requires socioeconomic principles that maintain this "circulation radius." On a temporal scale, the life spans of local resources should be also considered. For example, electricity must be consumed immediately after power generation, while woody biomass in forest stands can be saved (or grown) for the long term.

Therefore, utilizing forest resources is not enough to resolve forest and energy issues entirely in a modern society. That conclusion, however, is based on an assumption of unrestricted forest usage and energy consumption in an environment where only market competition exists. We have derived that it may be possible to construct a system that can provide solutions to biomass usage and forest issues according to two constraints described as follows:

[1] The geographic and temporal range that allows for resource circulation is limited.

[2] Facilities of feasible size within that range are limited.

These constraints were not explicit at the beginning of our research in local areas but rather were established more clearly as our research progressed.

4. Case Study: Mishima Town

4.1 Location and Forests

Oku-Aizu and Mishima in Fukushima Prefecture are snowy areas located in the southwest region of Aizu, close to the border with Niigata Prefecture (Fig. 2).

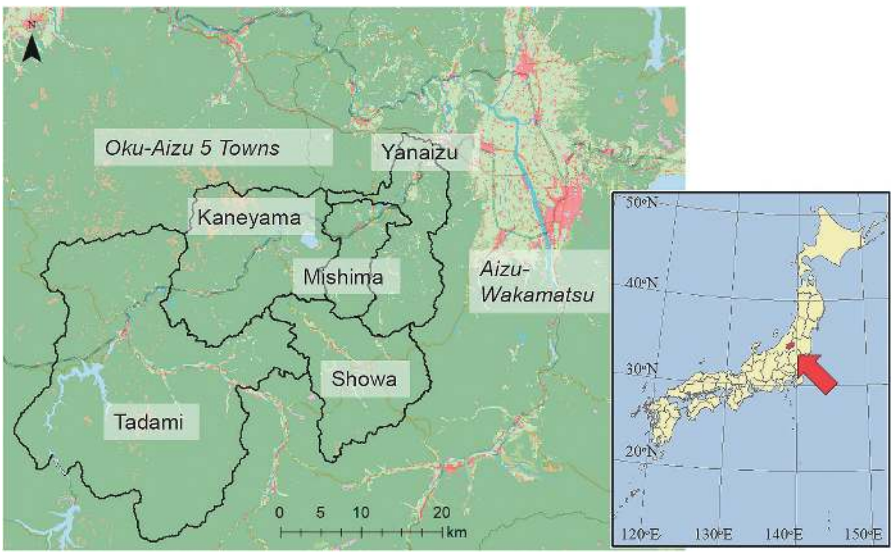

Fig. 2 Location of the Oku-Aizu region and Mishima Town.

According to the Automated Meteorological Data Acquisition System (AMeDAS) in Kaneyama (a town located next to Mishima), the average annual temperature in the area is 10.4°C, the annual rainfall is 2,020 mm, and the maximum snow depth is 177 cm. Because Mishima has a robust water supply due to snowmelt, the Tadami River has been used as a power source since the end of World War II. At that time, Mishima's population was over 7,000. Its current population, though, is less than 1,600, and Mishima exhibits one of the highest aging rates (53.0%) in Fukushima Prefecture.

The general environmental issues of Japan also affect Mishima, as forests cover 88% of its total area. While the town has a wealth of forest resources, cedar forests, which account for almost all of the privately-owned forest land, are often owned by small-scale owners. There is also a lack of clarity regarding the ownership of these lands. Timber prices have remained low for approximately 50 years, and not enough thinning or materials production have been conducted. Furthermore, the timber produced in the Aizu region presents issues, such as bending or excessive knots, rendering it a low-quality product in Japan.

The results of a questionnaire survey on forest-related awareness in Mishima (Nakamura & Togawa, 2020) showed that local people perceive that a combination of issues, such as the delineation of boundaries to subdivide forests and unclear ownership, leads to forest degradation. The details of this survey are discussed below.

Some areas of Mishima have adopted the Fukushima Forest Restoration Project (Fukushima Prefecture, 2018), which was implemented by the prefecture after the earthquake and nuclear accident. Future benefits from assessment of knowledge on forest boundaries through the project are expected.

4.2 Toward Renewable Energy from Forests

Aiming to promote forest management and profits from woody biomass, Mishima formulated the Mishima New Energy Vision Program in February 2006 and the Detailed Mishima New Energy Vision Program the following year. In particular, these programs proposed replacing petroleum-based boilers and stoves with units that can handle easy-to-use pellets as combustible sources to make use of the town's abundant forest resources. In cooperation with nearby towns and villages (Yanaizu, Kaneyama, Showa and Tadami), the Oku-Aizu Five-Towns Revitalization Council was established in April 2010 to achieve biomass utilization. However, the Tohoku Earthquake in 2011 and major damage from torrential rains in Niigata and Fukushima have contributed to decreased biomass processing.

The chambers of commerce of 13 cities, towns and villages in the Aizu area worked together to establish the Aizu 13 Project Council in 2016, with the participation of Mishima Town and its Chamber of Commerce. The goal of the project was to expand production of materials from the Aizu area, obtaining high-performance plywood materials, and additionally biomass as energy fuel sources. The project is currently being managed and actively implemented by both the Aizu Forest Resource Utilization Promotion Council and the Aizu Forest Utilization Organization, based on collaboration with the national governmental. Our project was affected indirectly by the Aizu 13 project, but it was not directly connected to the project.

4.3 Research Activities and Their Outputs

The Fukushima branch of the National Institute for Environmental Studies (NIES) holds surveys and performs studies which are assessed primarily by regional development researchers. It also conducts collaborative research with specialists from other universities and institutes. To ensure and promote regional research, NIES signed a collaborative agreement (memorandum of understanding) with Mishima Town in 2017.

With constraints [1] and [2] in mind, the research team listed problems in Mishima and the surrounding area, initiatives to solve those problems, the effects of the initiatives, and details regarding the research required to address those effects (Table 1).

Further details on the research activities in Table 1 are provided below.

4.3.1 Detailed Survey of Forest Resources, Questionnaire Survey on Forest-related Awareness

The first problem that arises in forest resource usage is that forest boundaries are frequently not well-defined or geographically accurate, as required for forestry practices. Furthermore, most forests are unmanaged, and the actual timber volume of the forest is unknown.

To reconfirm forest boundaries and estimate amounts of resources, we used a drone to take aerial images and laser measurements. We studied a forest area of approximately two hectares in a commonly owned forest in the Mishima area, contributing to understanding by the residents. The aerial images and laser measurements were taken in August 2019, whereas the field studies were conducted in February and May of 2020 (Fig. 3). With the cooperation of the Graduate School of Engineering at Osaka University, we used 3D-point cloud data to estimate tree top positions and crown sizes for each tree through machine learning. After its use in an initial analysis and validation, this technique continued to be improved for accuracy. These data show the amount of each type of tree located in each owner's forest and the available timber volume. In addition, the detailed aerial images and point cloud data may also be useful in defining boundaries of forest ownerships and planning forest practices. Data are currently being compiled for presentation to the residents of the town and the region.

Table 1 Problems and related initiatives in Mishima and its surrounding area.

Problem	Initiative	Effect	Secondary effect	Research contribution
(1) Forest degradation	Forest management	Forest consolidation, stabilized production, and reforestation	Disaster prevention, environmental conservation, and carbon sink development	A detailed survey of forest resources and questionnaire survey on forest-related awareness
(2) Decline of the forestry industry	Promotion of the forestry industry	Development of raw wood production with hi tech machineries	Revitalization of local demand and development of human resources	Estimation of potential production and analysis of the resource circulation area
(3) Reliance on external energy sources	Utilization of woody biomass	Introduction of small CHP, attracting demand for thermal energy	Provision of low-cost energy, energy security, and attraction of projects that use thermal energy	Energy conversion technology and a detailed survey of energy demand
(4) Economic decline of local communities	Industry, government, and private sector initiatives (establishment of councils and study groups)	Promotion of local cooperation	Revitalization of local communities/economies	Community system theory and research on future-community scenarios

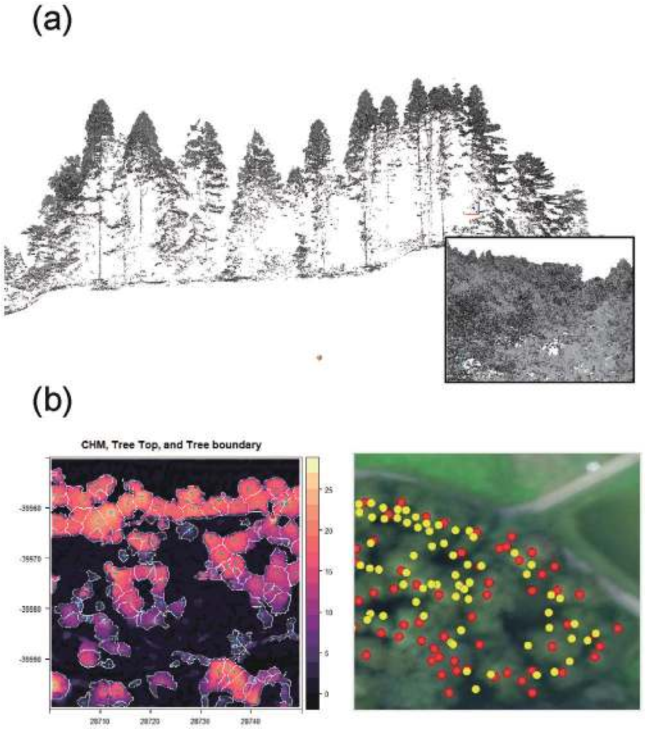

Fig. 3 (a) LiDAR image (cross section and overview) (b) Data processing images images (detection of tree crowns and tree tops).

This will also lead to higher consolidation and better forest land management.

Additionally, a questionnaire survey on forest-related awareness was conducted in 2018 (Nakamura & Togawa, 2020). We randomly selected 599 Mishima residents (aged 18 or older) and obtained responses from 264 people (44.1% response rate). The results showed that 76% of the respondents knew the location of the forest they owned, and only 58% were aware of its boundaries, indicating a need to reconfirm boundaries. However, because most owners did not manage their forests, when asked about owners' management difficulties, they mentioned "low timber prices" as the most frequent response, followed by "lack of manpower," "undeveloped work roads," and "unclear boundaries." Another common response was "aging of the population." When asked about subsidies they would need to keep managing their own forests, we obtained several answers related to financial issues (timber prices/supply chain: 56 responses, logging/transportation subsidies: 30 responses), as well as 80 responses related to boundary definition/forest road development.

This survey received significant attention from residents regarding disaster prevention and reduced forest functionality. Although the Great Tohoku Earthquake and nuclear disaster in 2011 had little impact on the region, torrential rain and flooding occurred in the same year. The relationship between forests, landslides and floods is understood well in this region, and biomass utilization plans should be considered.

4.3.2 Estimation of the Production Potential and Resource Circulation Area

A study on the biomass potential of privately owned forests in the Oku-Aizu area (Ooba et al., 2018) produced a cost analysis of chip production to obtain energy (including logging and transportation). It demonstrated that, for forestland with extremely low management rates (percent of area managed annually: 0.36% per year), chip production to obtain energy was not possible from an economic viewpoint. However, the study showed that it might be possible to produce chips at a cost of 7,000 JPY/m^3 or less (in a scenario where the amount harvested was increased by conducting extensive thinning) and that future production could be ramped up to approximately 10,000 m^3.

4.3.3 Energy Conversion Technology, a Detailed Survey on Energy Demand

The feasibility of implementing a biomass energy system was also studied at typical hot spring facilities in the region (Togawa et al., 2018). Based on data from a survey jointly conducted with Mishima, energy demand by season and time was estimated in facilities along the Tadami River. It was demonstrated that a small biomass co-generation system (operating continuously during all seasons) would consume approximately 700 tons/year of woody biomass. However, when the economics of this scenario were analyzed, there were concerns about profitability.

We installed Home Energy Management Systems (HEMS) in Mishima Town, which are currently studying the actual energy consumption of houses in the cold Oku-Aizu region and modeling that consumption. From these data, energy charts for each community in Mishima (Togawa et al., 2019) were prepared and efforts to estimate the distributed energy systems' potential in terms of demand were conducted (Fig. 4).

Based on this research, Mishima used a subsidy for promoting smart communities from Fukushima Prefecture to conduct a foundational study (pre-feasibility study, FS) in 2017 with a company, Nippon Koei Co., Ltd., on energy production for local demand using local resources, such as woody biomass in Mishima. In a study performed by a different welfare institution, a scenario in which electricity was sold to an electric company under the FIT system resulted in a profitable business and an estimated biomass consumption of approximately 800 tons/year.

4.3.4 Social System Theory and Research on Future Social Scenarios

Constraints [1] and [2], as well as the research described above, suggested that in Mishima, woody biomass materials can only be produced as fuel sources at a relatively small scale. This implies that the town needs a community system centered on introducing CHP systems with an electricity output of 50 kW or less. In addition, we may surmise that it would be realistic to start the project up relying on the current FIT system to some extent to ensure business revenues.

As a starting point for social implementation, in 2018, Mishima Town implemented small initiatives as a first step towards promoting woody biomass utilization. One of those initiatives was a Tree Station-style program, which has also been conducted by many municipalities and non-profit organizations, collecting firewood in December. In Mishima, 1 m^3 of logs can be exchanged for 4,500 JPY in local gift certificates. The heating and cooling system of the Mishima Local Crafts Museum, a core institution that shares the town's braided handicrafts tradition, was switched to a system based on a wood-fired boiler. This fuel is produced from firewood collected in Mishima using the method mentioned above.

The research team has not only promoted collaborative-scientific research in Mishima but has also held collaborative research meetings with civil engineering, architecture and regional policy officials at least once a month. Additionally, we have also provided our research results to the Mishima Town Forestry Policy Development Committee, through on-demand lectures in Mishima and public meetings that brought stakeholders together.

To address the implementation of an energy system, in 2019, the NIES proposed that Mishima establish a council. This council would divide tasks related to woody biomass utilization issues into forest/energy production-related (supply-related) and energy-consumption tasks, and form working groups with stakeholders comprised of both local and external parties that could consolidate the opinions of the higher-level coordination council and make decisions. Therefore, Mishima and four surrounding

Fig. 4 Visualization of the current and recommended energy systems in each settlement in Mishima Town (a) Heat and electricity demands (b) Analytical results: proposals for facilities and decreased CO_2 emissions.

towns established the Oku Aizu Five Towns and Villages Revitalization Council with a subsidy from the Ministry of the Environment. The council is currently making efforts to develop an energy system that relies on the use of municipal facilities.

5. Concluding Remarks

The Fifth Basic Environmental Plan approved by the Cabinet in 2018 focuses on the concept of "regional circular and ecological spheres" (regional CESs, MOE, 2020). Contributing to that plan, the Ministry of the Environment has elaborated several projects based on model cases of regional CESs. This concept can be considered an integrative version of conventional social-consciousness environmental goals, such as low-carbon or decarbonizing societies, recycling-oriented societies, and harmonious relationships with nature. The achievement of regional CESs infers the embodiment of the SDGs. Woody biomass usage, as presented by this project in Mishima and the Oku-Aizu region, has the potential to create a society that can achieve such goals (Fig. 5).

In addition to academic and technological support, this project will require public-private partnerships. Several businesses and non-profit organizations in Mishima are enthusiastic about renewable energy and woody biomass usage. Creating more effective and sustainable biomass, as well as regional economic circulation, will require higher logging than predicted from the energy center studies and FSs conducted so far. The amount of wood available for logging is limited, however, (and the energy demand is predicted to be significant). Local operators using local resources will conduct this biomass supply project; therefore, we would assume that the energy system the town might install could be a relatively small one and that the service area could be a district of Mishima (a community). Although it is small-scale, implementing this community revitalization energy project will not only ensure economic profits for the community, but also have definite secondary effects in terms of the development of industrial and human resources and stable local employment. This project would have a significant impact as a local service and could also be an initial step towards promoting future projects addressing a wider range of social services, similar to Germany's Stadtwerke project (Ando, 2019) and responding to the problems of depopulation and aging. This demonstrates the potential of the project to scale into a major sector in the rural regions.

We plan to continue our research on sustainable regions like regional CESs, ensuring that the local needs are considered through their potential environmental effects, which allows for an integrated approach. Furthermore, a thorough estimation of future full-scale biomass utilization is required, considering its impacts on the Oku-Aizu forests, where the vegetation has remained unchanged for several decades. We are also planning to begin research on adaptation to climate change in the Oku-Aizu region, including a survey aimed at creating hazard maps for landslides and other disasters, while actively providing information to local governments.

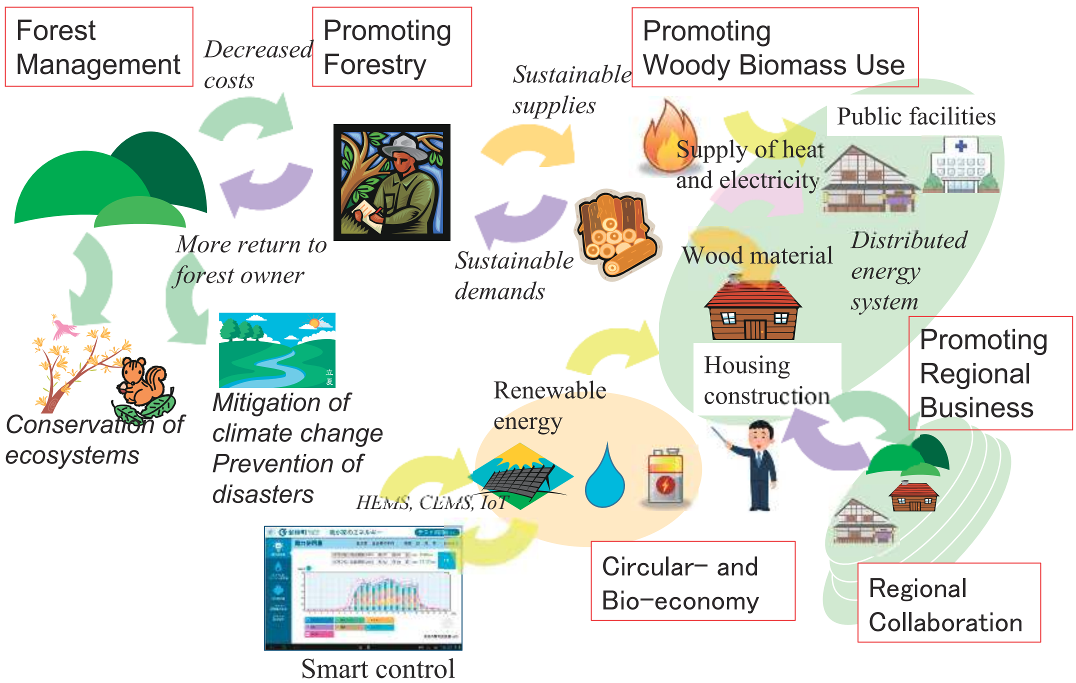

Fig. 5 Sustainable revitalization inside Oku-Aizu through the regional circulation of local resources from production to distribution and consumption as the goal of "regional circular and ecological spheres" (CESs).

References

Agency for Natural Resources and Energy (2019) *Guidebook for installing a distributed energy system for local production for local consumption considering regional characteristics*. (in Japanese) Retrieved from https://www.enecho.meti.go.jp/category/saving_and_new/bunsan_plat/02_03/pdf/002_04.pdf (Accessed 30 September 2020)

Amano, M. (2007) Development of evaluation model for carbon sink. *Abstract of the Final Report, Environment Research and Technology Development Fund of the Ministry of Environment, B-60*. Retrieved from https://www.env.go.jp/earth/suishinhi/wise/j/pdf/J06B0060Eng.pdf (Accessed 30 September 2020)

Ando, T. (2019) A study of municipal power producer and supplier projects and the ideal business style for local city. *Journal of Japan Solar Energy Society*, 45, 61–65.

Forestry Agency of Japan (2020) Forest environmental tax and forest environmental transfer tax. (in Japanese) Retrieved from https://www.rinya.maff.go.jp/j/keikaku/kankyouzei/kankyouzei_jouyozei.html (Accessed 30 September 2020)

Fukushima Prefecture (2018) Restoration project of forests inside Fukushima Prefecture. (in Japanese) Retrieved from https://www.pref.fukushima.lg.jp/site/portal/64-2.html (Accessed 30 September 2020)

Han, Q. and Chiba, Y. (2009) Leaf photosynthetic responses and related nitrogen changes associated with crown reclosure after thinning in a young *Chamaecyparis obtusa* stand. *Journal of Forest Research*, 14, 349–357.

Haruta, F. and Muraki, M. (2017) A study on possibility about separation of infrastructure and operation under district heating and cooling projects: A case study of decentralized energy infrastructure project. (in Japanese with English summary) *Journal of the City Planning Institute of Japan*, 52(3), 481–486. https://doi.org/10.11361/journalcpij.52.481

Hiroshima, T. and Nakajima, T. (2006) Estimation of sequestered carbon in Article-3.4 private planted forests in the first commitment period in Japan. *Journal of Forest Research*, 11, 427–437.

METI (The Ministry of Economy, Trade and Industry, Japan) (2017) Present status and promotion measures for the introduction of renewable energy in Japan. Retrieved from https://www.meti.go.jp/english/policy/energy_environment/renewable/index.html (Accessed 30 September 2020).

Miyata, S., Kosugi, K., Gomi, T. and Mizuyama, T. (2009) Effects of forest floor coverage on overland flow and soil erosion on hillslopes in Japanese cypress plantation forests. *Water Resources Research*, 45, W06402.

MOE (Ministry of Environment, Japan) (2020) Regional circular and ecological spheres. (in Japanese) Retrieved from https://www.env.go.jp/seisaku/list/kyoseiken/index.html (Accessed 30 September 2020)

Nakamura, S. and Togawa, T. (2020) T. Survey on household wood biomass use and energy consumption in the Oku-Aizu region. *Global Environmental Research*, 24, 237–242.

Ohnishi, S., Fujii, M., Fujita, T., Matsumoto, T., Dong, T., Akiyama, H. and Dong, H. (2016) Comparative analysis of recycling industry development in Japan following the Eco-Town program for eco-industrial development. *Journal of Cleaner Production*, 114, 95–102.

Ooba, M., Fujii, M., Togawa, T., Nakamura, S., Nemoto, K., Yi, D. and Tsuji, T. (2018) Simulation of effective carbon sequestration by promoting of woody biomass utilization from the upstream to downstream. *Proceedings of the Conference on Biomass Science*, 13, 15–16. (in Japanese with English summary) Retrieved from https://doi.org/10.20550/jiebiomassronbun. 13.0_15

Sakai, S., Tsuda, T. and Yasaka, M. (2017) Calculating the supply potential of unused wood for woody biomass power generation plants in Hokkaido, Japan. *Journal of the Japanese Forest Society*, 99, 233–240. (in Japanese with English summary) Retrieved from https://doi.org/10.4005/jjfs.99.233

Shima, N. and Nose, A. (2014) A feasibility study on of the heat and electricity supply installation using the gas cogeneration system. *2014 AKITA Technical Papers of Annual Meeting The Society of Heating, Air-Conditioning and Sanitary Engineers of Japan 2014*, 81–84. (in Japanese) Retrieved from https://doi.org/10.18948/shasetaikai.2014.2.0_81

Togawa, T., Yi, D., Ooba, M., Nakamura, S., Nemoto, K. and Fujii, M. (2019) Design of distributed energy system considering village structure and regional resource circulation in a mountainous area. *Proceedings of the Conference on Biomass Science*, 14, 19–20. (in Japanese with English summary) Retrieved from https://doi.org/10.20550/jiebiomassronbun.15.0_19

Togawa, T., Ooba, M., Nemoto, K. and Nakamura, S. (2018) Design and standards for biomass energy system considering with regional characteristics. *Proceedings of the Conference on Biomass Science*, 13, 43–44. (in Japanese with English summary) Retrieved from https://doi.org/10. 20550/jiebiomassronbun.13.0_43

Totman, C.D. (1995) *The Lumber Industry in Early Modern Japan*, University of Hawaii Press, Honolulu.

Uchiyama, Y. and Kohsaka, R. (2020) Utilization of forest environment transfer tax in ordinance-designated cities: trend of urban forest policy and its diversity in Japan. *Journal of the Japanese Forest Society*, 102, 173–179. Retrieved from https://doi.org/10.4005/jjfs.102.173

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